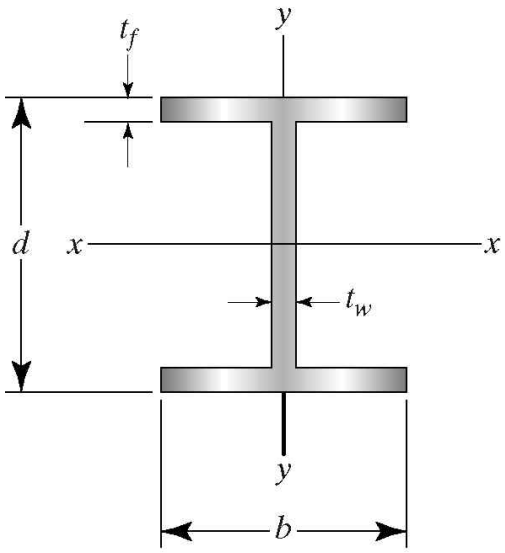
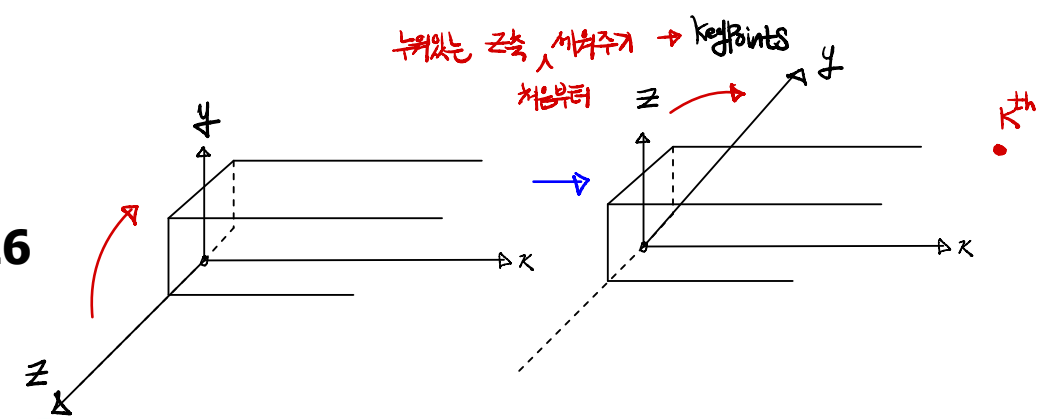


Appendix-D-1

Wide-flange sections or W shapes.

W12×26



Type: BEAM, Subtype: ASEC

Arbitrary: User-supplied integrated section properties instead of basic geometry data.

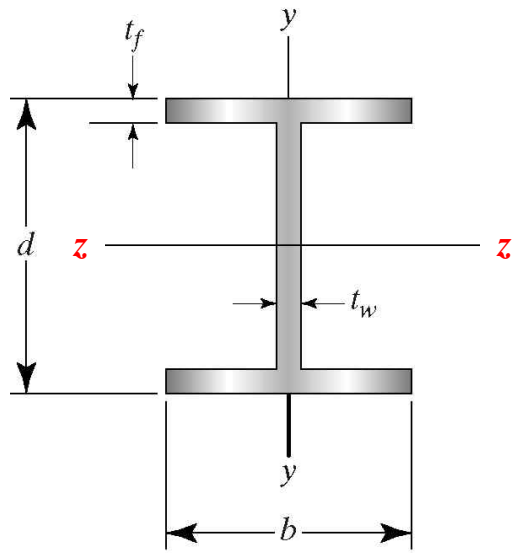
Data to provide in the value fields:

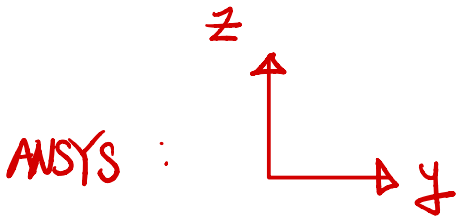
A, *Iyy*, *Iyz*, *Izz*, *Iw*, *J*, *CGy*, *CGz*, *SHy*, *SHz*, *TKz*, *TKy*

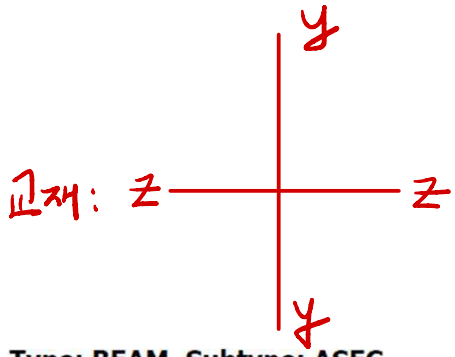
where

A = Area of section	7.65
Iyy = Moment of inertia about the y axis	17.3
Iyz = Product of inertia	0
Izz = Moment of inertia about the z axis	204
Iw = Warping constant	0
J = Torsional constant	0.3
CGy = y coordinate of centroid	0
CGz = z coordinate of centroid	0
SHy = y coordinate of shear center	0
SHz = z coordinate of shear center	0
TKz = Thickness along Z axis (maximum height)	6.49
TKy = Thickness along Y axis (maximum width)	12.22

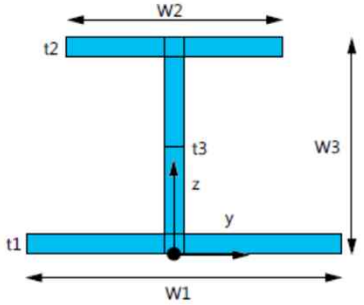
SECPLOT cannot display an ASEC plot.







Type: BEAM, Subtype: I



Data to provide in the value fields:

W1, *W2*, *W3*, *t1*, *t2*, *t3*

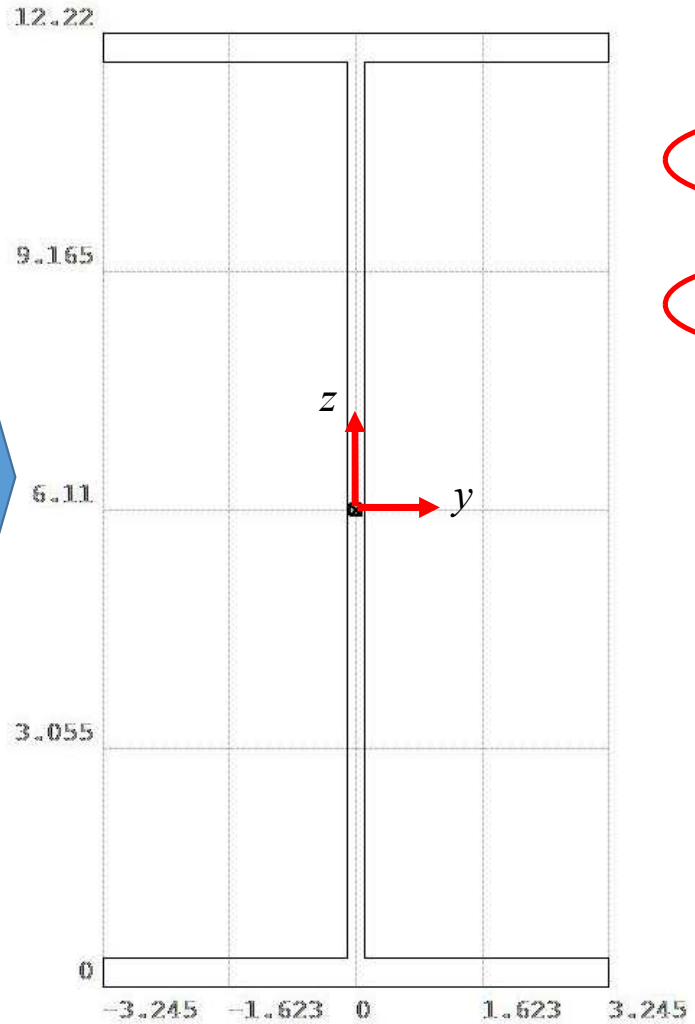
where

W1, *W2* = Width of the top and bottom flanges

W3 = Overall depth

t1, *t2* = Flange thicknesses

w1=w2=6.49
w3=12.22
t1=t2=0.380
t3=0.230




```
Area
   = 7.5602
Iyy
   = 201.769
Iyz
   = -.302E-13
Izz
   = 17.3244
Warping Constant
   = 605.419
Torsion Constant
   = .291315
Centroid Y
   = .411E-15
Centroid Z
   = 6.11
Shear Center Y
   = -.210E-13
Shear Center Z
   = 6.11
Shear Corr. YY
   = .56026
Shear Corr. YZ
   = -.220E-12
Shear Corr. ZZ
   = .352886
```

yy→zz

zz→yy

Type: BEAM, Subtype: ASEC

Arbitrary: User-supplied integrated section properties instead of basic geometry data.

Data to provide in the value fields:

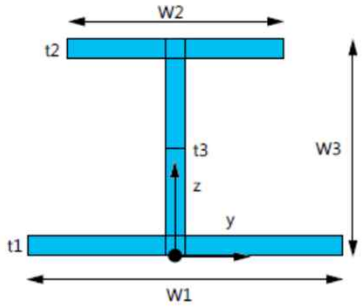
A, *Iyy*, *Iyz*, *Izz*, *Iw*, *J*, *CGy*, *CGz*, *SHy*, *SHz*, *TKz*, *TKy*

where

A = Area of section	7.65
Iyy = Moment of inertia about the y axis	17.3
Iyz = Product of inertia	0
Izz = Moment of inertia about the z axis	204
Iw = Warping constant	0
J = Torsional constant	0.3
CGy = y coordinate of centroid	0
CGz = z coordinate of centroid	0
SHy = y coordinate of shear center	0
SHz = z coordinate of shear center	0
TKz = Thickness along Z axis (maximum height)	6.49
TKy = Thickness along Y axis (maximum width)	12.22

SECPLOT cannot display an ASEC plot.

Type: BEAM, Subtype: I



Data to provide in the value fields:

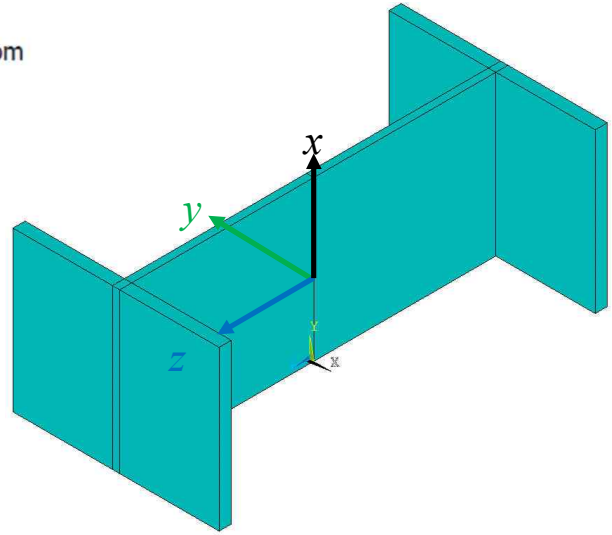
W1, *W2*, *W3*, *t1*, *t2*, *t3*

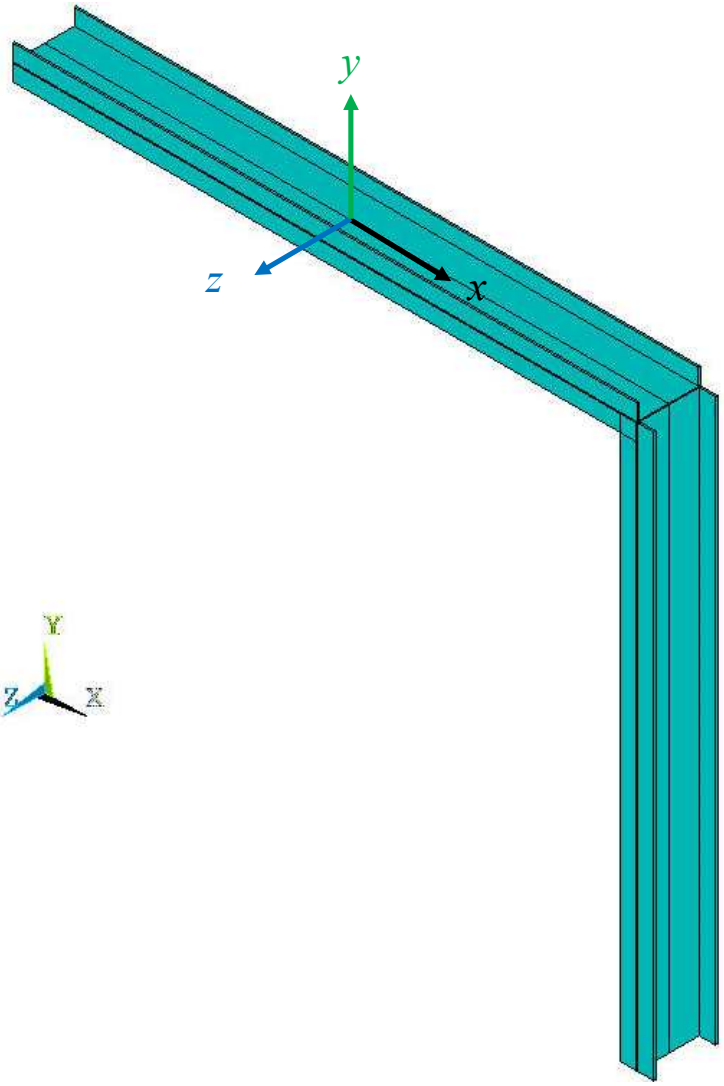
where

W1, *W2* = Width of the top and bottom flanges

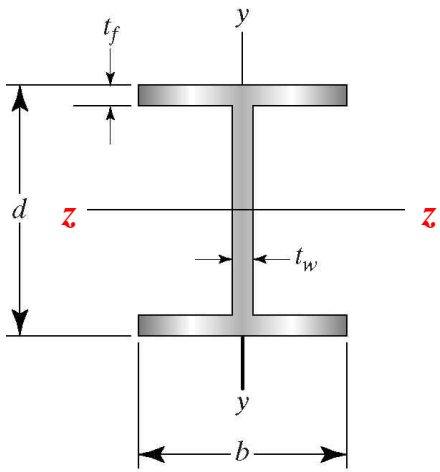
W3 = Overall depth

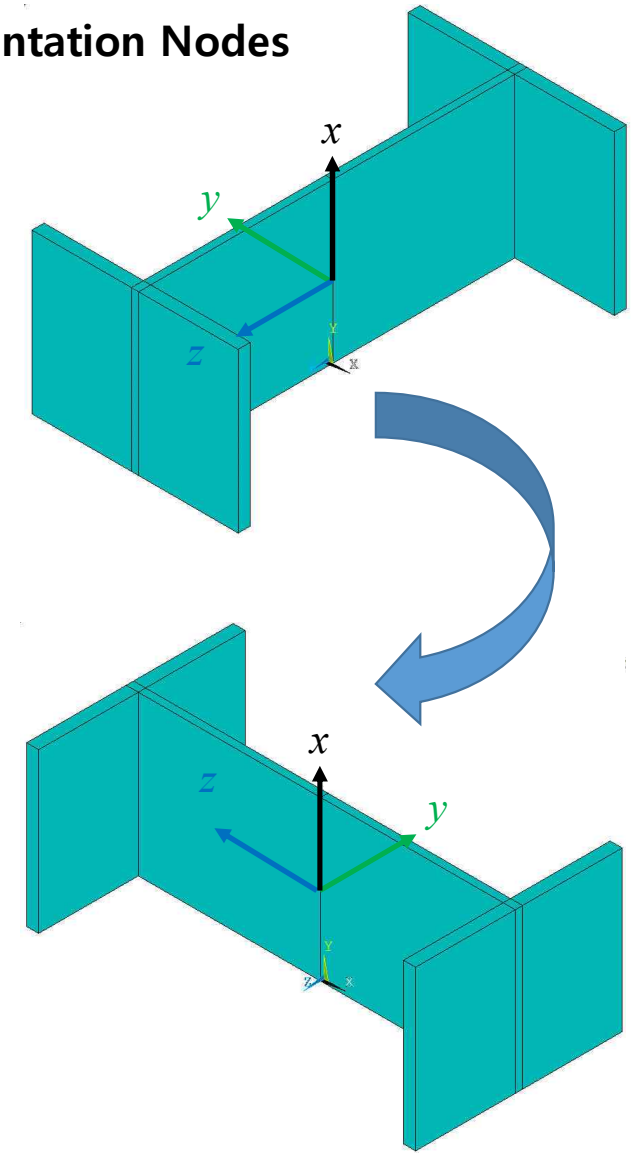
t1, *t2* = Flange thicknesses

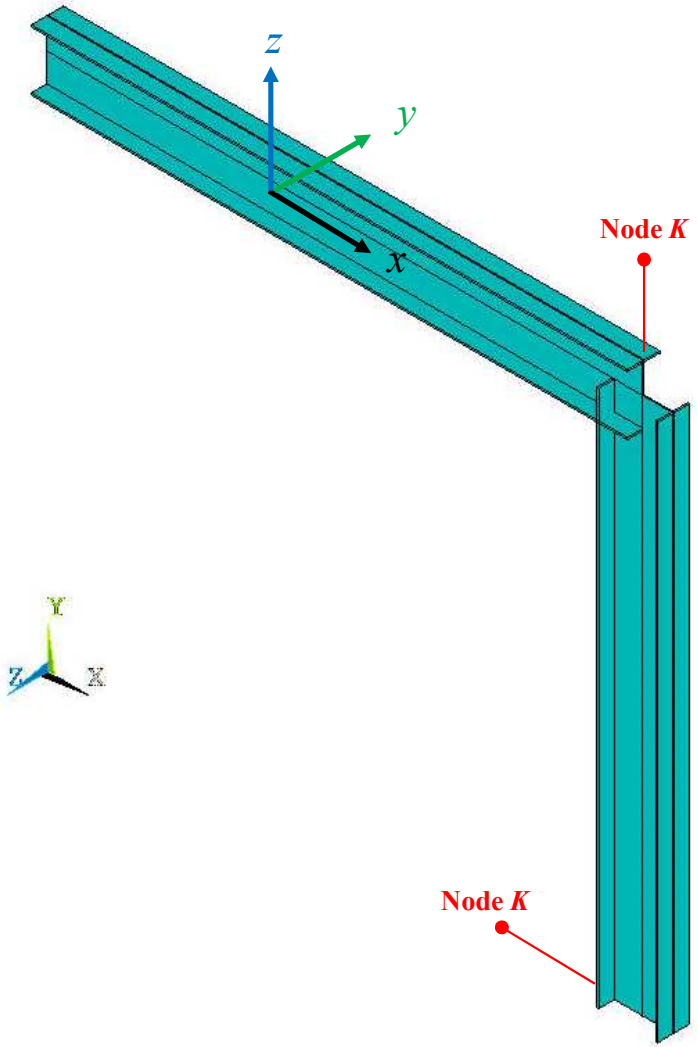


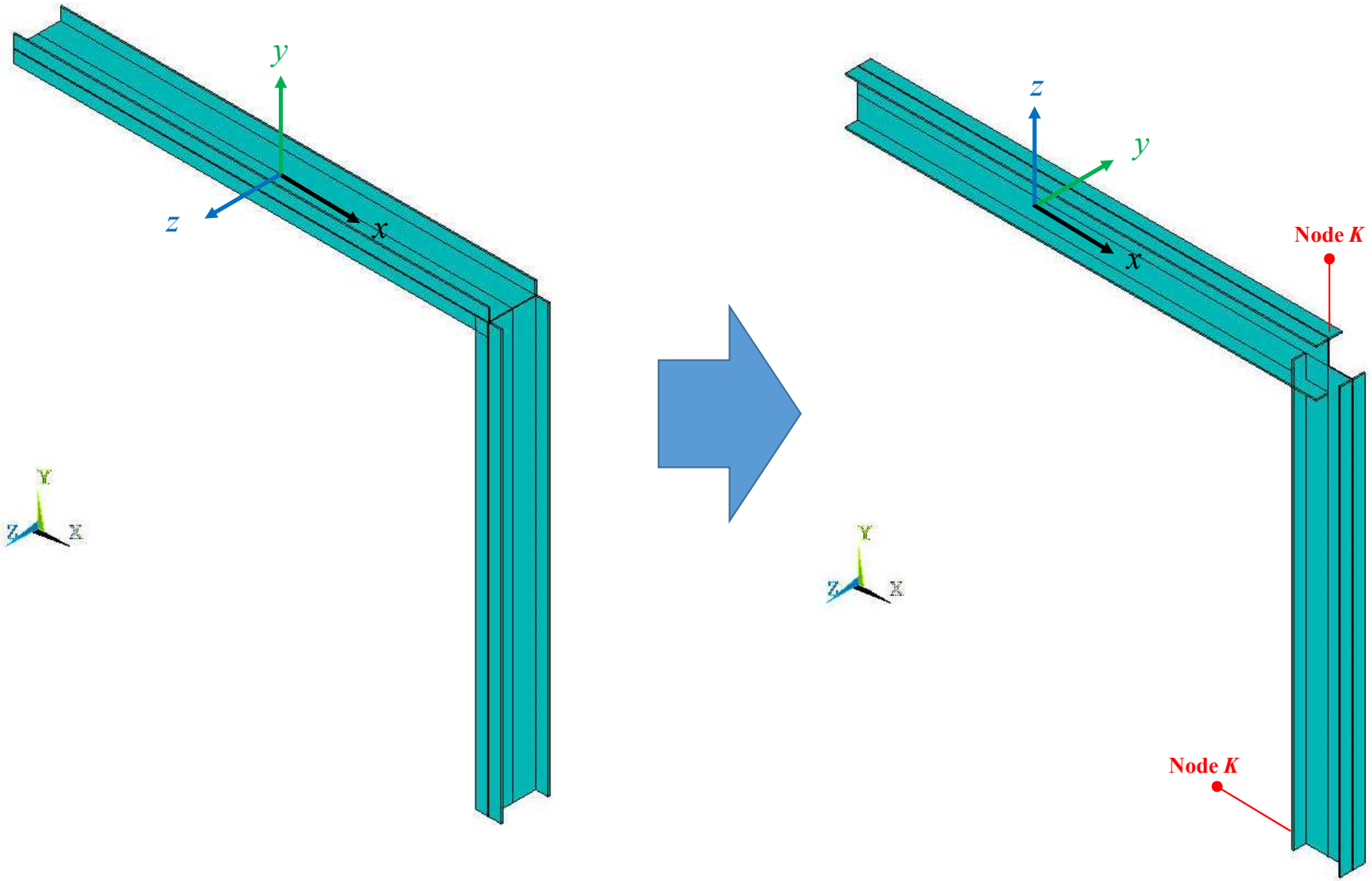


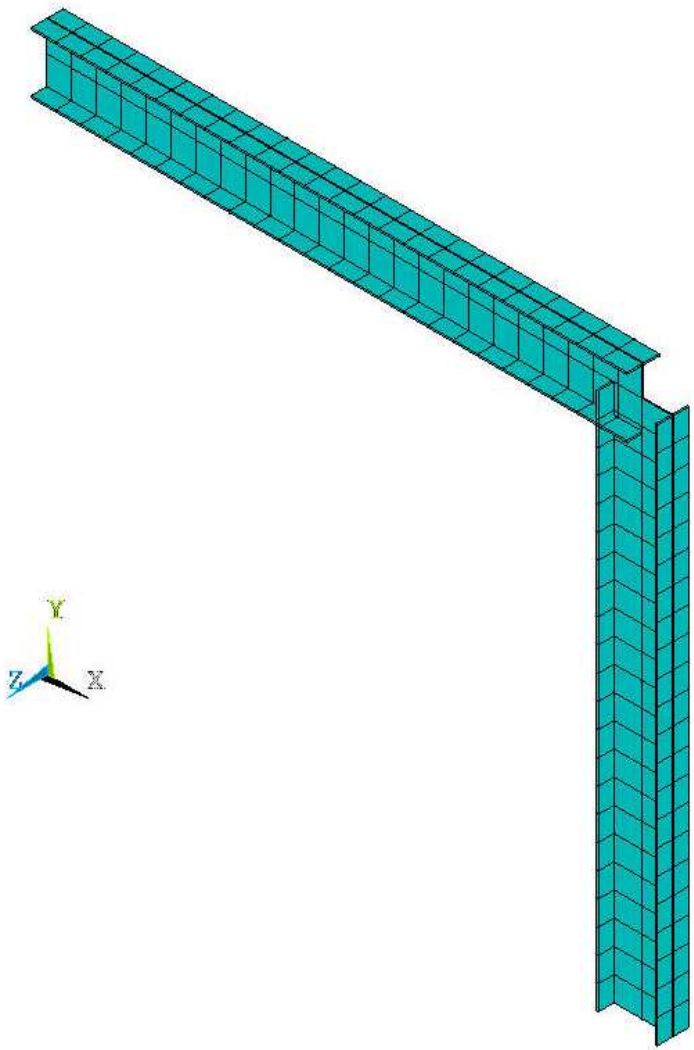
Beam Meshing With Orientation Nodes








y
z
x
Y
Z
X
z
y
x
Node K
Node K
Y
Z
X



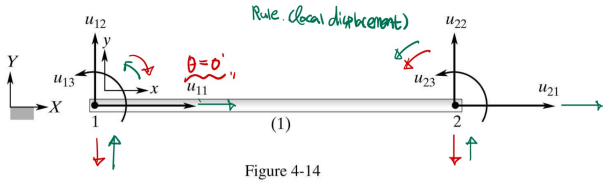
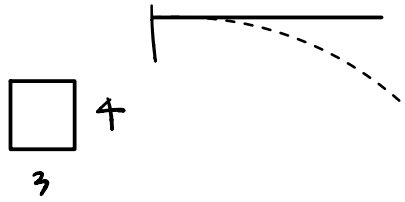
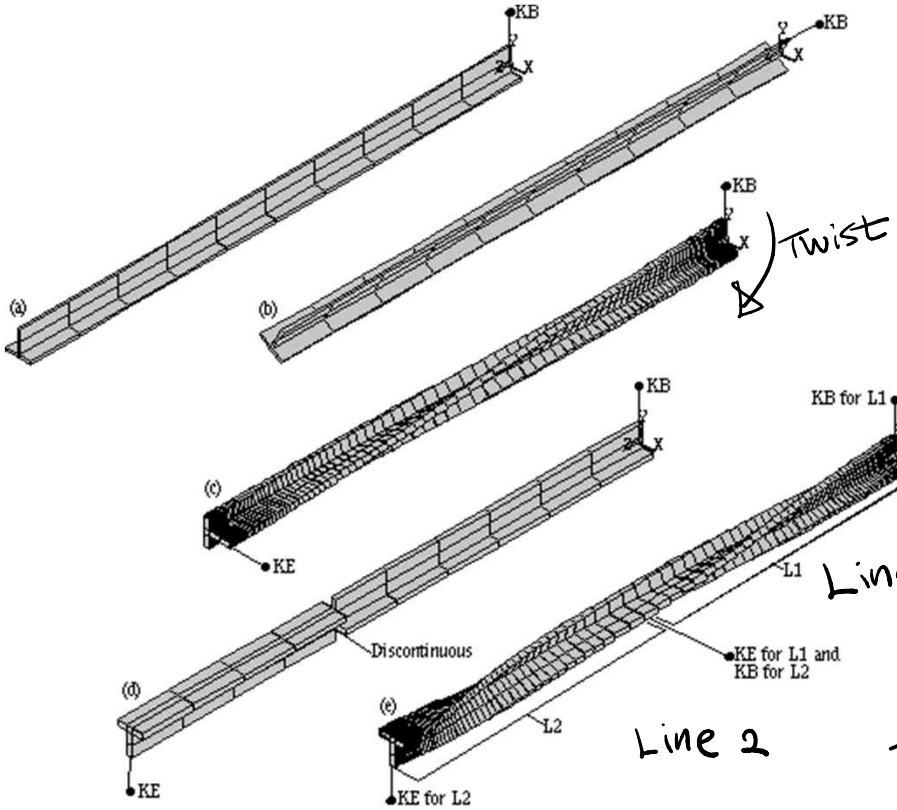


Figure 4-14
The configuration of element (1).



Beam Meshing with Orientation Keypoints

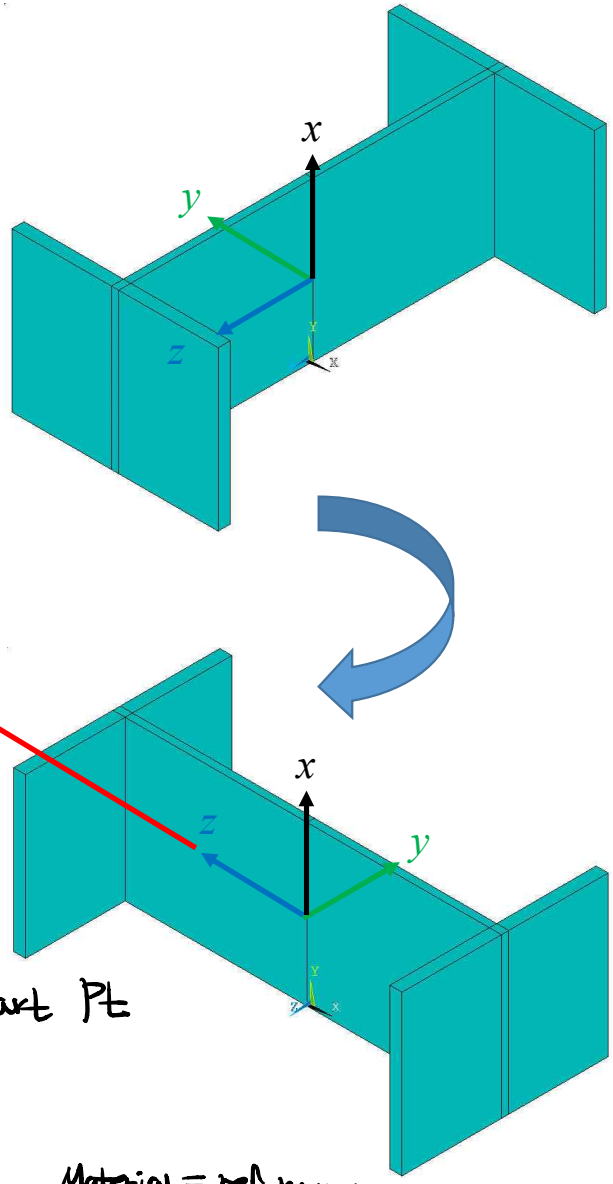


Line 1

Line 2

Line 2개.

L_1 End Pt = L_2 Start Pt



L, A, V

LATT, *MAT, REAL, TYPE, --, KB, KE, SECNUM*

Begin End.

Line Attribution -

: Keypts 생성 → Line 생성 → Line에 Element 생성 → Line's Element `LATT`로 할당.

- Material = ref num, 1
- TYPE = ref num, 1
- SECNUM = ref num, 1

→ `Lmesh` → Element 생성.
Line meshing.

KB로 고정해두기.

Chap. 4 Axial Members, Beams, And Frames

Example 4.5 An overhang frame supporting a distributed load

